The Weinberg Angle from Geometry

Electroweak Mixing Is 3/13

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Abstract

The Weinberg (weak mixing) angle determines electroweak unification. Its measured value $\sin^2(\text{theta_W}) = 0.2312$ has no derivation in the Standard Model. We show **$\sin^2(\text{theta_W}) = 3/(\text{GAUGE} + 1) = 3/13 = 0.2308$**, achieving **0.19% accuracy**. The numerator 3 = N_gen (fermion generations) and denominator 13 = GAUGE + 1 (gauge bosons + Higgs). The W and Z boson mass ratio follows directly.

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1. The Weinberg Angle

1.1 Definition

The Weinberg angle theta_W determines how the weak and electromagnetic forces mix:

- Photon: $A = \cos(\text{theta_W})\times B + \sin(\text{theta_W})\times W^3$
- Z boson: $Z = -\sin(\text{theta_W})\times B + \cos(\text{theta_W})\times W^3$

1.2 Measured Values

$\sin^2(\text{theta_W}) = 0.23121 \pm 0.00004$ (at Z pole)

$\cos(\text{theta_W}) = m_W/m_Z = 0.8815$

1.3 The Problem

In the Standard Model, $\sin^2(\text{theta_W})$ is a free parameter. Its value 0.23 is measured, not derived.

Grand Unified Theories predict $\sin^2(\text{theta_W}) = 3/8 = 0.375$ at the GUT scale, but this requires running down to the electroweak scale.

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2. The Z^2 Solution

2.1 The Formula

$\sin^2(\text{theta_W}) = \text{N_gen}/(\text{GAUGE} + 1)$

$\sin^2(\text{theta_W}) = 3/13 = 0.2308$

2.2 The Components

Component	Value	Meaning
N_gen	3	Fermion generations
GAUGE	12	Gauge generators
GAUGE + 1	13	With Higgs
Result	**3/13**	**$\sin^2(\text{theta_W})$**

2.3 Verification

Quantity	Predicted	Measured	Error
$\sin^2(\text{theta_W})$	0.23077	0.23121	**0.19%**
cos(theta_W)	0.8771	0.8815	0.50%
m_W/m_Z	0.8771	0.8815	0.50%

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3. Physical Interpretation

3.1 Why 3/13?

Numerator: 3 = N_gen - Three generations of fermions couple to the weak force - The weak interaction "sees" 3 generations

Denominator: 13 = GAUGE + 1 - 12 gauge bosons (8 gluons + 3 W + 1 B) - +1 for the Higgs boson - Total gauge + scalar content

The ratio: $\sin^2(\text{theta_W})$ = (matter content)/(force content) = 3/13

3.2 The Gauge Structure

GAUGE = 12 breaks down as: - SU(3): 8 gluons - SU(2): 3 W bosons (W^+, W^-, W^3) - U(1): 1 B boson

The +1 in the denominator is the Higgs, needed for mass generation.

3.3 Connection to GUT

Grand Unified SU(5) predicts: **$\sin^2(\theta_{W,GUT}) = 3/8 = 0.375$**

At electroweak scale: **$\sin^2(\theta_{W,EW}) = 3/13 = 0.2308$**

Both have **numerator 3** (generations). Denominators: 8 = CUBE vs 13 = GAUGE + 1

The running from GUT to EW scale: CUBE → GAUGE + 1 8 → 13 Δ = 5 = BEKENSTEIN + 1

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4. W and Z Boson Masses

4.1 The Mass Ratio

From $\sin^2(\theta_W) = 3/13$:

$\cos^2(\theta_W) = 1 - 3/13 = 10/13$

$\cos(\theta_W) = \sqrt{10/13} = 0.8771$

$m_W/m_Z = \cos(\theta_W) = 0.8771$

4.2 Comparison

Quantity	Predicted	Measured	Error
m_W/m_Z	0.8771	0.8815	0.50%

4.3 Absolute Masses

Using the Fermi constant G_F and fine structure constant alpha:

$m_W^2 = \pi \times \alpha / (\sqrt{2} \times G_F \times \sin^2(\theta_W))$

With Z^2 values: - $\alpha = 1/137.04$ - $\sin^2(\theta_W) = 3/13$

Result: - $m_W = 77.6$ GeV (vs measured 80.4 GeV) - $m_Z = 88.5$ GeV (vs measured 91.2 GeV)

Error: ~3% (from radiative corrections not included)

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5. The W-Z Mass Splitting

5.1 The Formula

Delta m = m_Z - m_W = m_Z × (1 - cos(theta_W))

= m_Z × (1 - sqrt(10/13))

5.2 Fractional Splitting

Delta m/m_Z = 1 - sqrt(10/13) = 0.1229 = 12.29%

Measured: 11.86%

5.3 Physical Meaning

The W-Z mass splitting is: **1 - sqrt(10/13) = 1 - sqrt((GAUGE + 1 - N_gen)/(GAUGE + 1))**

The splitting is determined by the ratio of weak content (3) to total content (13).

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6. The rho Parameter

6.1 Definition

The rho parameter tests the Standard Model structure:

rho = $m_W^2/(m_Z^2 \times \cos^2(\text{theta_W}))$

Tree level: rho = 1 (exactly)

6.2 From Z^2

With $\sin^2(\text{theta_W}) = 3/13$:

rho = 1 (exactly at tree level)

This confirms the SU(2) × U(1) structure is consistent with Z^2 geometry.

6.3 Measured Value

rho = 1.01061 ± 0.00020

The deviation from 1 is due to radiative corrections (top quark loops).

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7. Connection to Other Z^2 Results

7.1 The Web

The Weinberg angle connects to:

1. Gauge structure: GAUGE = $9Z^2/(8pi) = 12$

2. Three generations: N_gen = BEKENSTEIN - 1 = 3

3. Fine structure constant: $alpha^{-1} = 4Z^2 + 3 = 137.04$

4. Cabibbo angle: $sin^2(theta_c) = 1/(2 \times (GAUGE - 2)) = 1/20$

7.2 Unified Picture

All electroweak parameters derive from Z^2: - Mixing angle: 3/13 - Cabibbo angle: 1/sqrt(20) - Fine structure: 1/137.04

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8. Predictions

8.1 $sin^2(theta_W)$ Is Exact

$sin^2(theta_W) = 3/13 = 0.23076923\ldots$

Future precision should approach this value.

8.2 Running Interpretation

The Z^2 value 0.2308 matches the Z pole measurement.

This suggests Z^2 describes physics at the electroweak scale.

8.3 GUT Connection

The relation between GUT (3/8) and EW (3/13): - Same numerator (3 = generations) - Running changes denominator: 8 → 13

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9. Summary

The Problem

Why is $sin^2(theta_W) \approx 0.23$?

The Standard Model

Cannot explain it. It's a free parameter.

The Z^2 Answer

$\sin^2(\text{theta_W}) = \text{N_gen}/(\text{GAUGE} + 1) = 3/13$

Term	Value	Meaning
N_gen	3	Generations
GAUGE + 1	13	Gauge + Higgs
$\sin^2(\text{theta_W})$	**0.2308**	**0.19% from measured**

The Significance

Electroweak mixing is determined by: - 3 fermion generations - 13 gauge + scalar degrees of freedom - Ratio: 3/13 = 0.2308

The W and Z boson masses follow from this geometric mixing.

Appendix: Numerical Details

A.1 Z^2 Constant

$Z^2 = 32\text{pi}/3 = 33.5103$

A.2 GAUGE

$\text{GAUGE} = 9Z^2/(8\text{pi}) = 12$

A.3 N_gen

N_gen = BEKENSTEIN - 1 = 4 - 1 = 3

A.4 Weinberg Angle

$\sin^2(\text{theta_W}) = 3/(12 + 1) = 3/13 = 0.230769$

A.5 Comparison

Measured: 0.23121 Error: |0.23077 - 0.23121|/0.23121 = 0.19%

A.6 Mass Ratio

cos(theta_W) = sqrt(1 - 3/13) = sqrt(10/13) = 0.8771 m_W/m_Z (measured) = 80.38/91.19 = 0.8815 Error: 0.50%

"Electroweak mixing is 3 parts weak to 10 parts electromagnetic."

— Carl Zimmerman, 2026

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Repository: https://github.com/carlzimmerman/zimmerman-formula